Aarya Patil

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Research Interests

My research lies at the intersection of astrophysics, statistics, and computer science. I develop and apply novel data-driven methods to understand how galaxies like our own Milky Way form and evolve.

Education

2023 **Direct-Entry PhD in Astronomy & Astrophysics**, University of Toronto — Canada
Thesis title: Order in Chaos - Decoding the age-metallicity structure of the Milky Way disk
Supervisors: Jo Bovy & Gwendolyn Eadie

2018 **BEng in Computer Engineering**, S. P. Pune University, (CGPA: 9.45/10) — India

Positions

Aug 2026 - **Assistant Professor of Statistics, Astronomy & Astrophysics**, Penn State
Institute for Computational and Data Sciences (ICDS) Co-Hire, ICDS AI Hub Advisory Committee

2023 - 2026 **LSST Discovery Alliance Catalyst Fellow**, MPIA — USD 100,000+/year
Independent fellowship with USD 15k research budget/year

2021 - **Elected Governance Committee Member (Lead Role)**, The Astropy Project
Coordination Committee (since 2025): Responsible for overall project management
Finance Committee (since 2021): Acquisition/allocation of external funding totaling ∼3.5M USD

2017 **Developer**, Google Summer of Code (GSoC) Participant with OpenAstronomy/Astropy
Mentors: T. Aldcroft (Harvard-Smithsonian), M. van Kerkwijk (UToronto) & H. M. Günther (MIT)

Selected Awards & Honors

2026 - 2027 **MPIA Postdoctoral Fellowship** (catalyst fellowship extension, declined) — Euro 90,000

2025 **Lancelot M. Berkeley–N.Y. Community Trust Prize for The Astropy Project**
Annual prize for meritorious work in astronomy by the American Astronomical Society (AAS)

2024, 2022 **Astrostatistics Student Paper Competition Finalist** (top 5) — USD 100/year
American Statistical Association/Astrostatistics Interest Group (ASA/AIG)

2023 **IOP Publishing Top Cited Paper Award for The Astropy Project**

2023 - 2025 **Schmidt AI in Science Postdoctoral Fellowship** (declined) — USD 78,000/year

2022 **ADASS Software Prize for The Astropy Project**

2022 - 2023 **Data Sciences Institute Doctoral Student Fellowship** — CAD 25,000/year
Data Sciences Institute, University of Toronto, up to 3 years of funding

2021 - 2022 **International Graduate Student Doctoral Fellowship** — CAD 3,000
Astronomy & Astrophysics, University of Toronto, academic excellence award

2018 - 2023 **University of Toronto Graduate Fellowships** — CAD 23,250/year + tuition

2018 - 2021 **Massey College Junior Fellowship, University of Toronto (UofT)** — CAD 11,000

2017 **ABU ROBOCON - All India Rank 10** out of 125 teams, programming head of team

Research Grants (incl. telescope time)

2026 **James Webb Space Telescope Cycle 5 Archival or Theory Research proposal**
Highly Parallelized Grism Spectroscopy for Stellar & Extragalactic Astronomy

2025 **COST (European Cooperation in Science and Technology) Action** Euro 125-150k/year
Management Committee Germany rep. for 4-year Action: Testing Fundamental Physics with Seismology

2025 **International Astronomical Union Hands-On Workshops** (conditional) Euro 29,200
Co-Investigator (Co-I): "Learning Data Science for Exploring Astronomical Archives" workshop in Kenya
Awarded but unable to run school due to lack of additional funds

Astropy Cycle III/IV Funding Grants USD 20,860
2024 Lead: Pan-African School for Emerging Astronomers (PASEA) 2024 in Tunisia (USD 640)
2023 Co-I: Python/Astropy Training School in Bulgaria (USD 8,670)
2022 Lead: PASEA 2022 in Zambia (USD 11,550)

Dunlap Institute (UofT) Seed Funding Grants CAD 35,968
2022 Co-I: PASEA 2022 in Zambia (CAD 29,100)
2022 Co-I: Intertwining Dunlap/UofT & Sustainable Open-Source Software via Astropy (CAD 6,868)

Refereed Publications

5 first-author, 2 second-author with mentored students, 2 Astropy collaboration papers (∼14,500 citations).

Published or in press:

2026 **Patil, A. A.**; Aerts, C.; Wang, N. Y. N., Van Beeck, J. & Pedersen, M. G. "Beyond prewhitening: Detection of gravity modes and their period spacings in Slowly Pulsating B stars using the multitaper F-test". Astronomy & Astrophysics, Volume 714, article A234, 17 pp. [arXiv/2512.10019]

2026 de Souza, R. S. et al. incl. **Patil, A. A.** for the COIN Collaboration; "sagui: SED-based segmentation of multiband Galaxy images – application to JADES in GOODS-South". Monthly Notices of the Royal Astronomical Society, Volume 549, Issue 4, stag1062 [arXiv/2604.18812]

2026 Dage, K. C. et al. incl. **Patil, A. A.**; "The Secret Lives of Open Clusters: a Multiwavelength Examination of Three Open Clusters". Publications of the Astronomical Society of Australia, Volume 43, id.e013, 10 pp. [arXiv/2601.08087]

2025 **Patil, A. A.**; Eadie, G.; Speagle, J. & Thomson, D. "Improving Harmonic Analysis using Multitapering: Precise frequency estimation of stellar oscillations using the harmonic F-test". The Astronomical Journal, Volume 170, Issue 1, id.7, 13pp. [arXiv/2405.18509] [6 citations]

2025 Sun, J.*; **Patil, A.**; Li, Y.; Guo, J. L. C. & Zhou, S. "Collaboration Challenges and Opportunities in Developing Scientific Open-Source Software Ecosystems: A Case Study on Astropy". Proceedings of the ACM on Human-Computer Interaction, Volume 9, No. 7, article CSCW281 (November 2025), 33pp. [doi: 10.1145/3757462] [1]. *student

2025 Sun, J.*; **Patil, A.**; Li, Y.; Guo, J. L. C. & Zhou, S. "Advancing Sustainable Communities in Scientific OSS: A Replication Study with Astropy". 2025 IEEE/ACM 18th International Conference on Cooperative and Human Aspects of Software Engineering (CHASE), Ottawa, ON, Canada, 2025, pp. 161-172, [doi: 10.1109/CHASE66643.2025.00027]. *student

[1] In computer science/engineering, papers in peer-reviewed conferences are as high-quality as (or more than) in journals.

2024 **Patil, A. A.**; Eadie, G.; Speagle, J. & Thomson, D. "Improving Power Spectrum Estimation Using Multitapering: Efficient Asteroseismic Analyses for Understanding Stars, the Milky Way, and Beyond". The Astronomical Journal, Volume 168, Issue 5, article id. 193, 21pp. [arXiv/2209.15027][9 citations]

2023 **Patil, A. A.**; Bovy, J.; Jaimungal, S.; Frankel, N. & Leung, H. W. "Decoding the age-chemical structure of the Milky Way disc: an application of copulas and elicitable maps". Monthly Notices of the Royal Astronomical Society, Volume 526, Issue 2, pp. 1997-2016 [arXiv/2306.09319] [12 citations]

2022 **Patil, A. A.**; Bovy, J.; Eadie, G. & Jaimungal, S. "Functional Data Analysis for Extracting the Intrinsic Dimensionality of Spectra: Application to Chemical Homogeneity in the Open Cluster M67". The Astrophysical Journal, Volume 926, Issue 1, article id. 51, 24pp. [arXiv/2109.10891] [8 citations]

2022 The Astropy Collaboration, Price-Whelan, A. M.; Lim, P. L.; Earl, N.; Starkman, N.; Bradley, L.; Shupe, D. L.; **Patil, A. A.** et al. "The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package". The Astrophysical Journal, Volume 935, Issue 2, article id 167, 20pp. [arXiv/2206.14220] [5947 citations]

2018 The Astropy Collaboration et al. incl. **Patil, A. A.**. "The Astropy Project: Building an Open-science Project and Status of the v2.0 Core Package". The Astronomical Journal, Volume 156, Issue 3, article id. 123, 19pp. [arXiv/1801.02634] [11130 citations]

In preparation:

Patil A. A.; Rix, H.-W. et al. "Euclid: Building a foundational stellar inventory using slitless infrared spectroscopy", To be submitted to the Euclid consortium for internal review by Nov 2026.

Wang, N. Y. N.*; **Patil, A. A.** & Aerts, C. "Estimating the lifetime of gravity modes in Slowly Pulsating B stars". To be submitted to Astronomy & Astrophysics by Dec 2026. *student

Non-refereed

2022 **Patil, A.** "aaryapatil/tapify: v0.1.0". Zenodo. 10.5281/zenodo.7312220

2022 Cruz, K.; Günther, H. M.; **Patil, A.**; Swinbank, J.; & Tollerud, E. "Astropy Proposal for Enhancement 19: Distributing Astropy Project Funding (APE19)". Technical Report, Zenodo. doi.org/10.5281/zenodo.6312048

2021 Robitaille, T. et al. incl. **Patil, A.** "astropy/astropy: v4.2.1", Zenodo. doi.org/10.5281/zenodo.4670729

2020 **Patil, A.**; Bovy, J.; & Eadie G. "Likelihood-free Inference of Chemical Homogeneity in Open Clusters". Joint Statistical Meetings Proceedings, ASA, pp 1838-1844.

Supervision & Mentorship

2026 - Present **Thinh Quy Doan**, Thesis Committee Member, PhD in Statistics, Penn State

2026 - Present **Chloe Klare**, Thesis Committee Member, PhD in Astronomy & Astrophysics, Penn State

2024 - Present **Nikki Yat Ning Wang** (BASc graduate, UCL → MSc Physics, Technion)
MPIA summer internship: Time-series analysis to probe the interior of massive stars

2024 - 2025 **Jenny Su** (PhD, Astronomy & Astrophysics, UofT) with Gwendolyn Eadie (UofT)
Estimating the periodicities of RR Lyrae stars using the multitaper F-test

2024 - 2025 **Rohith Pudari** (PhD, Computer Eng., UofT) with Shurui Zhou (UofT)
Large Language Model (LLM) to generate tutorials for scientific software

2022 - Present **Jiayi Sun** (PhD, Computer Eng., UofT) with Shurui Zhou (UofT) and Jin Guo (McGill)
Improving collaboration efficiency of open-source scientific software teams

Summer 2021 **Suyog Garg** (BTech, IIITDM → MSc Physics, UTokyo) with Hans Moritz Guenther (MIT)
GSoC project: "Implementing the MRT/CDS table standards in Astropy"

2019 - 2020 **Stephanie Joachim** (High school, Harbord Collegiate Institute, Toronto)
Massey College Tutoring and Mentorship Program for students in need of academic support

Teaching

Instructor (materials available on GitHub)

2027 **STAT 300: Statistical Modeling I** Penn State
Spring semester course on linear and logistic regression, analysis of variance, modeling for real data in R

Pan-African School for Emerging Astronomers (PASEA)

2024 Co-designed/taught (team of 3) a week-long (40 hrs) postgraduate data science course Monastir, Tunisia

2022 Similar duties plus helped develop/teach a one-day (8 hrs) undergraduate course Livingstone, Zambia

2023 **The Astropy training school** Sofia, Bulgaria
Co-designed/taught (team of 4) a week-long (40 hrs) Astropy course for Eastern European students/scientists

2022 **PASEA Alumni Research Program** (virtual)
Co-designed/taught (team of 3) a week-long (40 hrs) astronomical data analysis course for PASEA alumni

Guest lecturer

2025 **Kapteyn Astronomical Institute**, 14 March Groningen, Netherlands
Designed/taught a 1 hr time-series analysis lecture for advanced undergraduate/graduate astronomy students

2024 **International Max-Planck Research School Student Symposium**, 15 Nov. MPA, Garching
Delivered a 'Milky Way and smaller scales' lecture for PhD students; provide feedback on their presentations

Teaching Assistant (TA)

Winter 2021, 22 **Head TA**, AST 201: The Sun and its Neighbours, UofT
Led 30+ TAs for a 1000+ student course for non-science/engineering majors

Fall 2020, 21 **TA**, AST 221: Stars and Planets, UofT
Designed/ran weekly tutorials, held office hours, graded exams for astrophysics majors

2018 - 2020 **TA**, AST 201 and AST 101: Stars and Galaxies, UofT
Ran tutorials (40 students of 1000), planetarium shows, observing nights; designed test questions

Workshops

2026 **Statistical Thinking for Scientists, Time Series and Gaussian Processes** MPIA, Germany
Designed and taught 12+ hours of hands-on data science workshops using GitHub Copilot AI for coding

2025 **Euclid workshop: resolved stellar populations in/around the Galaxy** Leiden, Netherlands
Ran a hands-on tutorial on Euclid DR1 Data and Science, focusing on spectroscopy

2024 **2024 Astropy SURF Tutorial & Hackathon** Utrecht, Netherlands
Co-organized a hack day and ran Python/Astropy tutorial sessions for university students across Netherlands

2024 **Introduction to Statistics 101** MPIA, Germany
Designed and taught 8+ hours of hands-on statistics workshops with coding applications

Training & Certification

2021 - 2022 Teaching Fundamentals Certificate, Teaching Assistants' Training Program, UofT

2020 Advanced Training in Academic Writing and Speaking, GCAC, UofT

Selected Presentations (2020 - Present)

Invited Conference/Press Talks

2025 **LSST Discovery Alliance Catalyst Symposium**, Nov. 19 — Tucson, USA
Career Pathways Panelist and Plenary talk

2025 **Rubin First Look Watch Party in Heidelberg**, June 23
Partnering in Discovery: MPIA and Rubin (virtual)

2025 **Astropy coordination meeting**, June 23 — Santa Cruz, USA
State of the Astropy Project

2024 **LSST Discovery Alliance Catalyst Symposium**, Oct. 21 — Chicago, USA
Plenary talk: Bridging Astrophysics & Data Science

2024 **243rd meeting of the AAS**, Jan. 8 — New Orleans, USA
Expert panelist: Building on 25 Years of Community Organization in Astro Software Development

2023 **LSST Discovery Alliance Catalyst Symposium**, Oct. 23 — Tucson, USA
Building a unified model of the Milky Way galaxy using Rubin/LSST

2022 **Astronomical Software Development Workshop**, May 20 — New York, USA
Project Governance & Management Session Lead (format modified due to COVID-19)

2021 **Statistical Challenges in Modern Astronomy VII Conference**, June 10
Bayesian Breakout: Likelihood-free Inference of Chemical Homogeneity in Open Clusters (virtual)

2020 **Joint Conf. for Sch. & Uni. students on Natural & Math. Sciences, Ukraine**, Dec. 3
How did the Milky Way Galaxy Form and Evolve? (virtual)

Invited Colloquia & Seminars

2026 **University of Florida Colloquium**, Oct 20 — Gainesville, Florida
A Billion Stars, One Galaxy: Reconstructing the Milky Way's History with Astrostatistics

2026 **CAStAi lunch seminar**, Oct 8 — Penn State, USA
Millions of Spectra, Many Surveys, One Inference Problem: Statistical Learning at Survey Scale

2026 **Leiden Observatory Colloquium**, June 8 — Leiden, Netherlands
Astrostatistical innovations to decode the history of our Galaxy

2026 **Instituto de Estudios Astrofísicos Talk for the Milky Way Group** — Santiago, Chile
Galactic archaeology using LSST

2025 **Resolved Stellar Populations Workshop in the era of Euclid**, Dec. 2 — Leiden, Netherlands
Euclid stellar spectroscopy

2025 **Florida State University Astrophysics Group Seminar**, Nov. 13
Improving time-series analysis for time-domain astronomy

2025 **Heidelberg Institute for Theoretical Studies guest speaker**, Nov. 4 — Heidelberg, Germany
Beyond prewhitening: Improving time-series analysis for massive star asteroseismology

2025 **Euclid SWG-MWRSP teleconference**, March 20
Euclid stellar spectroscopy (virtual)

2025 **Kapteyn Astronomical Institute Seminar**, March 13 — Groningen, Netherlands
Decoding the history of our Galaxy

2025 **Institute of Astronomy (IvS) Seminar**, March 6 — Leuven, Belgium
Improved time-series analysis in the golden era of asteroseismology

2024 **Tartu Observatory Weekly Astronomical Seminar**, Oct 1 — Tartu, Estonia
Asteroseismology to understand stars and the Milky Way

2024 **KIPAC Tea Talk**, Aug 9 — Stanford, USA
Improving Time-Series Analysis to understand stars, the Milky Way, and beyond

2024 **LSST Discovery Alliance Institutional Members Meeting**, March 13
Community Impact: Data Science with LSST workshops (virtual)

2024 **Florida State University Astrophysics Group Seminar**, Feb. 21
Building a unified model of the Milky Way galaxy using astrostatistics (virtual)

2024 **Königstuhl Colloquium**, Feb. 16 — Heidelberg, Germany
Understanding the formation history of the Milky Way galaxy using astrostatistics

2023 **Toronto AstrophysicS Talks, Y'all (TASTY)**, Nov. 2023 — Toronto, Canada
Order in Chaos: Decoding the Age-Metallicity Structure of the Milky Way disk

2023 **NRC Herzberg Astronomy and Astrophysics Seminar**, March 1 — Victoria, Canada
Building a unified model of the Milky Way galaxy

2022 **Good Vibrations Seminar**, Oct. 26
Multitaper Spectral Analysis: Precise asteroseismic modeling of stars, exoplanets, and beyond (virtual)

2021 **Statistics & Machine Learning (SMILE) Journal Club, UofT**, Nov. 19
"Multitaper Spectral Estimation for Asteroseismology" (virtual)

2021 **Women of Aeronautics and Astronautics (WoAA), India**, Dec. 4
Around the World Speaker Series: From Computer Engineering to Astrophysics (virtual)

2020 **International CHASC AstroStatistics Centre, Harvard University**, Nov. 17
Likelihood-free Inference of Chemical Homogeneity in Open Clusters (virtual)

2020 **SMILE, UofT**, October 16
Introduction to Neural Networks (virtual)

Contributed Conferences Talks

2026	Euclid consortium meeting: MWRSP science working group splinter	Barcelona, Spain
2026	Astroinformatics 2026	Cologne, Germany
2025	ML4Astro conference	Heidelberg, Germany
2024	LSST@Europe6	La Palma, Spain
2020 - 2024	Joint Statistical Meetings (JSM) Conference	USA/Canada
2024	Astropy coordination meeting	Utrecht, Netherlands
2024	243rd meeting of the AAS	New Orleans, USA
2023	Astrostatistics in Canada and Beyond	Banff, Canada
2022, 2023	TASC/KASC Workshop	USA/Belgium
2023	Canadian Astronomical Society (CASCA) Meeting	Penticton, Canada
2023	Wide-Field Spectroscopy vs Galaxy Formation Theory	Tucson, USA
2022	Multitaper Spectral Analysis Workshop	virtual
2021	HRMOS Science Workshop	virtual
2020, 2021	Sloan Digital Sky Survey Meeting	virtual
2021	GALactic Archaeology with HERMES Science Meeting	virtual
2021	Stellar Stats Workshop, UofT	virtual

Selected Professional Activities & Services

2026 **Invited NSF Proposal Review Panelist**

2025 **Cosmostatistics Initiative Network (COIN) Member**
Part of the COIN residence program in Brazil working on astrostatistics and astroinformatics

2024 - 2025 **Program Chair**, ASA/Astrostatistics Interest Group
Organized astrostatistics invited/contributed sessions at the Joint Statistical Meetings 2025
Helped run the group's annual Student Paper Competition

2024 - Present **Manuscript Referee**, Journal of the American Statistical Association, Digital Signal Processing, RAS Techniques and Instruments, Astronomy & Astrophysics, AAS Journals, JOSS

Workshops & Schools

2027 (future) **Data Science in Astronomy School**, Lead Organizer — India
Proposal to organize a school at the Inter-University Centre for Astronomy and Astrophysics

2026 **Astropy Coordination Meeting**, Scientific Organizing Committee (SOC) — UK

2025 **AstroStatistics and Research-Oriented Data Analysis**, SOC — UK
Helped organise the ASTRODAT workshop at Durham University in September 2025

2022, 2024 **PASEA**, Co-organizer — Zambia, Tunisia
Refer to the PASEA paper: Strubbe et al. (2021), Nature Astronomy, Volume 5, p. 217-220
Two PASEA co-directors won the 2024 IAU Astronomy Education Prize

2022 **Gaia Hike Workshop**, SOC — Canada
Co-developed the talk/tutorial schedule and led the unconference session programming

Service at Home Institution

Penn State

2026 - Present **AI Hub Advisory Committee**, ICDS

2026 - Present **Graduate Student Advisee**, Department of Statistics

2026 - Present **Recruitment & Website Committee**, Department of Astronomy & Astrophysics

Max Planck Institute for Astronomy

2023 - Present **Fundamental Skills Workshop Co-organizer**, Data Science Group
Organize and run monthly workshops on data science skills as part of the group

2024 **MPIA Summer Internship Committee Member**
Propose internship projects, review applications, conduct interviews, and coordinate the internship

University of Toronto

2022 - 2023 **SMILE Journal Club Co-organizer**, Astronomy & Astrophysics
Inviting local/external speakers in astrostatistics and astroinformatics; organizing talks

2021 - 2023 **Learn Astropy Project Local Representative**, Dunlap Institute
Improving the computing skills of the Institute and developing educational resources for Astropy

2020 - 2023 **Governing Board Risk Committee Member**, Massey College
First Elected Student on the Governing Board; helped develop the COVID-19 risk plan

2019 - 2020 **Treasurer of Student Life, MasseyScope Outreach Co-founder**, Massey College

2018 - 2023 **Outreach Volunteer**, AstroTours, Astronomy on Tap, Planet Party, Science Rendezvous

2018 - 2021 **Mental Health, Health & Safety, Course Committee Member**, Grad. Astronomy

References

Prof. Hans-Walter Rix
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Director of the Galaxies and Cosmology Department
Max Planck Institute for Astronomy, Germany

Prof. Jo Bovy
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Chair, Professor of Astronomy & Astrophysics
University of Toronto, Canada

Prof. Conny Aerts
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Professor of Physics & Astronomy
KU Leuven, Belgium

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